Aim:

To implement the Cyclic Redundancy Check (CRC) method for detecting errors in data transmission over a communication channel.

Description:

Cyclic Redundancy Check (CRC) is an error-detection technique used in computer networks. The sender divides the data by a predefined generator polynomial using modulo-2 (XOR) division. The remainder obtained is called the CRC and is appended to the original data before transmission.

At the receiver, the received data is again divided by the same generator polynomial.

- If the remainder is 0, the data is assumed to be error-free.
- If the remainder is non-zero, an error is detected.

Code:

```
#include <iostream>
#include <string>
using namespace std;
string crcDivide(string data, string polynomial) {
    int n = polynomial.length();
    for (int i = 0; i <= data.length() - n; i++) {
        if (data[i] == '1') {
            for (int j = 0; j < n; j++) {
                data[i + j] =
                    (data[i + j] == polynomial[j]) ? '0' : '1';      } }}
    return data.substr(data.length() - n + 1);}
    int main() {
    string data, polynomial;
    cout << "Enter data: ";
    cin >> data;
```

```
    cout << "Enter polynomial: ";
    cin >> polynomial;

    string temp = data + string(polynomial.length() - 1, '0');

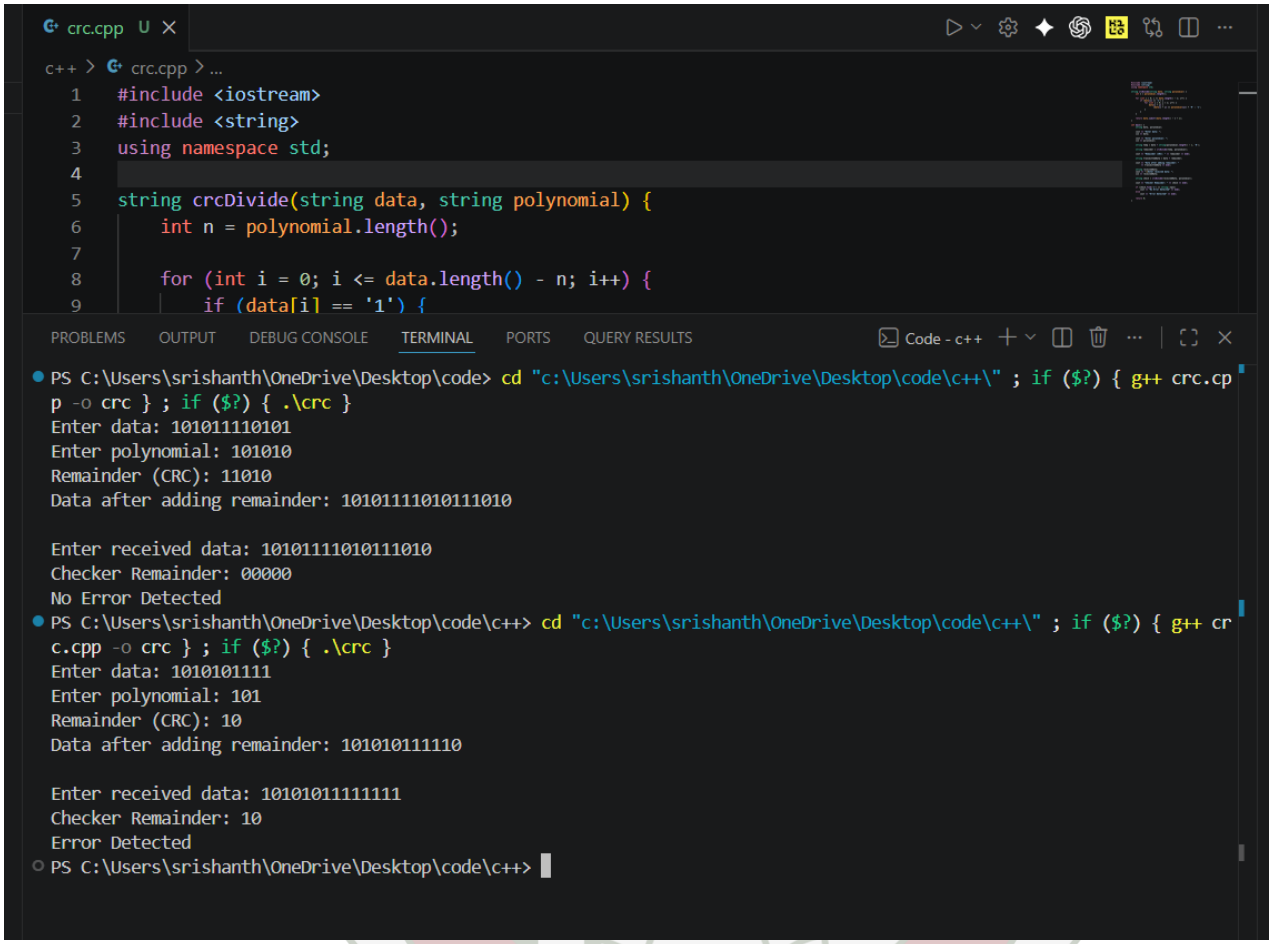
    string remainder = crcDivide(temp, polynomial);

    cout << "Remainder (CRC): " << remainder << endl;

    string transmittedData = data + remainder;
    cout << "Data after adding remainder: "
         << transmittedData << endl;
    string receivedData;
    cout << "\nEnter received data: ";
    cin >> receivedData;
    string check = crcDivide(receivedData, polynomial);
    cout << "Checker Remainder: " << check << endl;
    if (check.find('1') == string::npos)
        cout << "No Error Detected" << endl;
    else
        cout << "Error Detected" << endl;

    return 0;
}
```

Output:



Justification:

CRC is implemented because it provides fast and reliable error detection, especially for burst errors in transmitted data. It is widely used in computer networks and communication systems because it requires relatively simple hardware/software implementation and is more effective than simple parity checking.